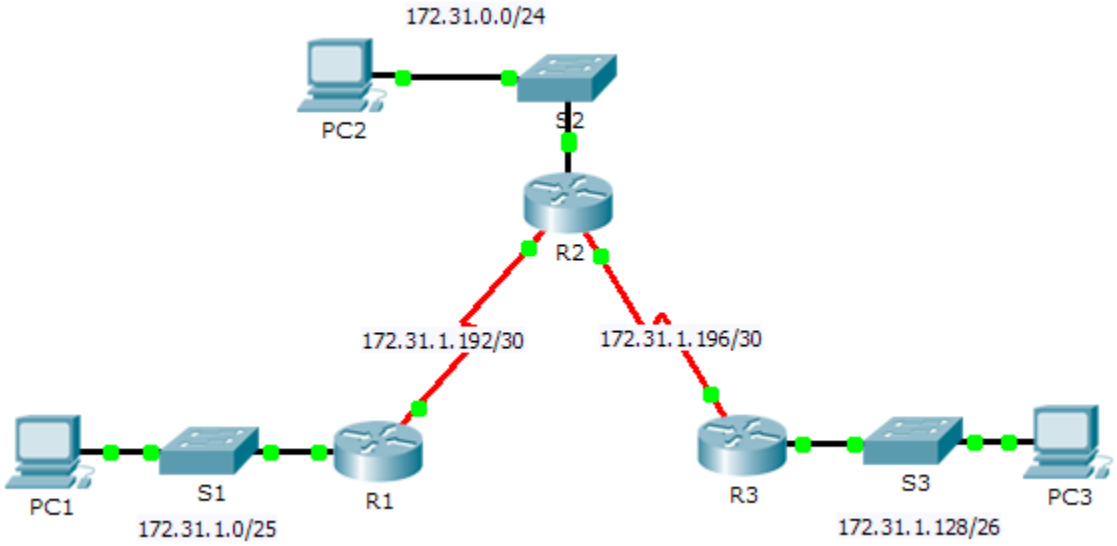
Packet Tracer - Configuring IPv4 Static and Default Routes

Topology



Addressing Table

Device	Interface	IPv4 Address	Subnet Mask	Default Gateway
R1	G0/0	172.31.1.1	255.255.255.128	N/A
	S0/0/0	172.31.1.194	255.255.255.252	N/A
R2	G0/0	172.31.0.1	255.255.255.0	N/A
	S0/0/0	172.31.1.193	255.255.255.252	N/A
	S0/0/1	172.31.1.197	255.255.255.252	N/A
R3	G0/0	172.31.1.129	255.255.255.192	N/A
	S0/0/1	172.31.1.198	255.255.255.252	N/A
PC1	NIC	172.31.1.126	255.255.255.128	172.31.1.1
PC2	NIC	172.31.0.254	255.255.255.0	172.31.0.1
PC3	NIC	172.31.1.190	255.255.255.192	172.31.1.129

Objectives

Part 1: Examine the Network and Evaluate the Need for Static Routing

Part 2: Configure Static and Default Routes

Part 3: Verify Connectivity

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Background

In this activity, you will configure static and default routes. A static route is a route that is entered manually by the network administrator to create a reliable and safe route. There are four different static routes that are used in this activity: a recursive static route, a directly attached static route, a fully specified static route, and a default route.

Part 1: Examine the Network and Evaluate the Need for Static Routing

a. Looking at the topology diagram, how many networks are there in total? 5

b. How many networks are directly connected to R1, R2, and R3? 2,3,2

c. How many static routes are required by each router to reach networks that are not directly connected?
8

d. Test connectivity to the R2 and R3 LANs by pinging PC2 and PC3 from PC1.

Why were you unsuccessful? All PC's are located in different remote network and cannot be pinged.

Part 2: Configure Static and Default Routes

Step 1: Configure recursive static routes on R1.

a. What is recursive static route?

A static route that requires another routing entry to resolve the next-hop IP and select the outgoing interface.

b. Why does a recursive static route require two routing table lookups?

A recursive static route requires two routing table lookups because the router must first find the route to the next-hop IP address. Then, it performs a second lookup to determine the actual outgoing interface used to reach that next hop.

c. Configure a recursive static route to every network not directly connected to R1, including the WAN link between R2 and R3.

d. Test connectivity to the R2 LAN and ping the IP addresses of PC2 and PC3.

Why were you unsuccessful?

Ping replies were not received from PC2 and PC3.

Step 2: Configure directly attached static routes on R2.

a. How does a directly attached static route differ from a recursive static route?

A directly attached static route specifies the outgoing interface directly, so the router forwards packets without any additional lookup. In contrast, a recursive static route specifies only the next-hop IP address, requiring an extra routing table lookup to determine the outgoing interface.

b. Configure a directly attached static route from R2 to every network not directly connected.

c. Which command only displays directly connected networks? show ip route

d. Which command only displays the static routes listed in the routing table? show ip route static

e. When viewing the entire routing table, how can you distinguish between a directly attached static route and a directly connected network?

By their route codes and details in the routing table: a directly attached static route is marked with S and shows an outgoing interface, while a directly connected network is marked with C and indicates a network

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Step 3: Configure a default route on R3.

a. How does a default route differ from a regular static route?

A default route is used when no more specific route matches a destination and typically points to 0.0.0.0/0, acting as a gateway of last resort. A regular static route is configured for a specific destination network and is used only for traffic matching that network.

b. Configure a default route on R3 so that every network not directly connected is reachable.

c. How is a static route displayed in the routing table? A static route is displayed in the routing table with the

Step 4: Document the commands for fully specified routes.

Note: Packet Tracer does not currently support configuring fully specified static routes. Therefore, in this step, document the configuration for fully specified routes.

a. Explain a fully specified route.

A fully specified route is a static route that explicitly includes both the outgoing interface and the next-hop IP address, so the router does not need to perform a recursive lookup to forward packets.

b. Which command provides a fully specified static route from R3 to the R2 LAN?

ip route 172.31.1.196 255.255.255.252 s0/0/1 172.31.1.197

c. Write a fully specified route from R3 to the network between R2 and R1. Do not configure the route; just calculate it.

ip route 172.31.1.192 255.255.255.252 s0/0/1 172.31.1.197

d. Write a fully specified static route from R3 to the R1 LAN. Do not configure the route; just calculate it.

ip route 172.31.1.0 255.255.255.128 s0/0/1 172.31.1.197

Step 5: Verify static route configurations.

Use the appropriate **show** commands to verify correct configurations.

Which **show** commands can you use to verify that the static routes are configured correctly?

show ip route

Part 3: Verify Connectivity

Every device should now be able to ping every other device. If not, review your static and default route configurations.

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Suggested Scoring Rubric

Activity Section	Question Location	Possible Points	Earned Points
Part 1: Examine the Network and Evaluate the Need for Static Routing	a - d	10	
	Part 1 Total	**10**	
Part 2: Configure Static and Default Routes	Step 1	7	
	Step 2	7	
	Step 3	3	
	Step 4	10	
	Step 5	3	
	Part 2 Total	**30**	
	Packet Tracer Score	**60**	
	Total Score	**100**	

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